Kush Dasadia

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Product builder at the intersection of AI and physical products.

EDUCATION

University of California, Berkeley · M.Eng, Mechanical Engineering — Aug 2024 – May 2025

Concentration: Controls of Robotics & Autonomous Systems.

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat · B.Tech, Mechanical Engineering — Jul 2019 – Jun 2023

EXPERIENCE

Founder · Fable Engineering — Apr 2025 – present · San Francisco

Operating company for hardware-first, AI-native product ventures. Pre-seed from The House Fund. Built and shipped five product lines across smart wearables, social robotics, kids' companions, and AI workflows (see Projects).

- **Hardware:** industrial design, 3D-printed prototype iteration loops, ODM sourcing at CES and in China, partnerships with European industrial design studios.
- **AI agent infrastructure:** voice + vision agent harnesses, planner / executor patterns, retrieval over personal context, structured eval harnesses, on-device vs. cloud split.
- **User research at scale:** 500+ interviews across creative professionals, students, robot-vacuum owners, and parents; synthesized findings into go / kill product decisions.
- **GTM:** LinkedIn beta launches, Reddit community building, 200+ outbound contacts, parallel customer discovery across multiple ICPs.
- **Fundraising:** closed Fable Engineering's pre-seed round (The House Fund); previously raised $250K pre-seed for Companion Robots on the strength of a working droid prototype.

Robotics Engineer · MPC Lab, UC Berkeley (Prof. Francesco Borrelli) — Aug 2024 – May 2025 · Berkeley, CA

- Designed modular in-field morphing solutions for a 1/10-scale autonomous vehicle (US Navy capstone) using goosenecks to enhance performance in dynamic environments.
- Collaborated on control challenges and rapid design modifications across vehicle dynamics, materials engineering, 3D printing, and control theory.

Sr. Drilling Officer (Company Man) · Sun Petrochemicals, India — Jul 2023 – Jun 2024 · India

- Managed a 100+ engineer team across drilling, mud, cementing, and HSE functions on both onshore and offshore rigs (well depths 4,000–4,300 ft); owned end-to-end drilling projects from planning and equipment selection to fluid management, casing, and cementing operations.
- Coordinated rig contractors, mud engineers, cementing crews, vendors, and HSE leads; ran daily safety, performance, and cost reporting; held vendor selection, procurement, and budget accountability for active well programs.

Research Intern · Institute of Automotive Technology, TU Munich — Feb 2022 – Aug 2022 · Remote

- Conceptualized an automated charging system for battery-electric trucks; designed for compatibility, efficiency, and user-friendliness across multiple OEM platforms.
- Analyzed the heavy-duty EV charging landscape: emerging standards (CCS, MCS), depot vs en-route topologies, and integration trade-offs with grid and fleet operations.

PROJECTS

Selected ventures built at Fable Engineering. Each is a distinct product line with its own thesis, prototype, and outcomes.

Super8 · Fable Engineering · Product, Engineering, GTM · super8.so

AI product visualization with 250+ users and generating revenue. Infinite-canvas, node-based agent that compiles natural-language intent into CAD / Blender / image-gen workflows.

- Designed the node-based canvas interface and the planner / executor agent; structured eval harness on fal.ai serverless infra (cold starts, shared pools, async submit-and-poll).
- Ran GTM across a LinkedIn beta launch, Reddit communities, and a 200-person outbound list; parallel customer discovery across growth marketers and industrial designers.

Murmur · Fable Engineering · Full-stack hardware + software, User research
Voice-enabled smart ring + AR-glasses bundle for ambient capture and contextual surfacing.

- **Traction:** 1,200+ waitlist; working prototype shipped to soft-launch users; AR-glasses + ring bundle live as Batch 01 reservations.
- **Hardware:** designed titanium enclosure; ran 3D-printed iteration loops; sourced ODMs at CES and in China; Belgian studio partnership on Gen-2 enclosure.
- **Project management:** drove timelines from prototyping through pilot runs; aligned mechanical design with firmware bring-up and system integration milestones.
- **Hands-on builds:** assembled mechanical components, fixtures, and electronic modules across iterations; from 3D-printed prototype through ODM pilot.
- **Full stack:** ring firmware (BLE, audio, touch, power); backend (transcription, retrieval, async LLM); voice agent (planner / executor); iOS app with integrations across Notion, Calendar, Gmail, Slack, ChatGPT.

Zero · Fable Engineering · Hardware, Firmware, Industrial design
Kid's desktop companion robot, built end-to-end in a 72-hour sprint.

- Hardware: Raspberry Pi 5 compute; 4.2" E-Ink face; dual speakers, camera, microphone, USB-C battery.
- Sprint methodology: Day 0 mood boards → Day 1 sketch → AI rendering → Fusion 360 CAD → overnight 3D print → Day 2 electronics + firmware bring-up → Day 3 vision + conversation integration.
- Event-driven Python service stack; lightweight on-device inference + remote LLM / VLM calls with guardrails.

Sonic · Fable Engineering · Hardware, Mechanical design, Robot learning
Low-cost semi-humanoid robotics platform built for AI / manipulation research and experimentation.

- Designed the torso frame and structural assemblies in CAD; 3D-printed parts for rapid iteration cycles.
- Selected off-the-shelf actuators across joints to keep BOM low while preserving DOF for manipulation experiments.
- Integrated and ran the π_0 vision-language-action foundation model (Physical Intelligence) on the platform; designed and conducted manipulation and behavior experiments.
- Worked with Unitree G1 humanoid for teleoperation and data collection.
- Implemented whole-body control VR teleop methods such as SONIC and TWIST.

Formula Student (FSAE) · SVNIT · Powertrain Lead, Electric & Combustion vehicles
Powertrain mechanical design and integration for Formula SAE race cars across both IC and EV classes.

- **IC powertrain (mechanical):** engine mounting and vibration isolation within the space-frame; intake design (restrictor, plenum, runner length) for target torque curve; exhaust header geometry (4-1 / 4-2-1) with routing and decibel compliance; fuel system layout (tank, lines, rail); cooling system (radiator placement, hose routing, expansion).
- **EV powertrain (mechanical):** motor mounting and torque-reaction structures; accumulator enclosure design (cell-stack packaging, structural rigidity, crash protection per FSAE EV rules); inverter mounting and cooling jacket; high-voltage cable routing and shielding; BMS packaging and serviceability.
- **Drivetrain integration:** torque path from engine / motor to wheels; chain-drive geometry, sprocket selection, and tensioning; differential mounting integrated with rear bulkhead; drivetrain packaging within chassis envelope; on-track validation and refinement.

SKILLS

AI & agents	Voice + vision agents, planner / executor patterns, retrieval over personal context (RAG), foundation models (LLM / VLM / VLA — π_0), structured evals, on-device vs. cloud split, fal.ai serverless.
Robotics	Humanoid platforms (Unitree G1), VR teleoperation & data collection, whole-body control (SONIC, TWIST), foundation-model integration on hardware, manipulation & behavior experiments, vehicle dynamics & controls.
Hardware	Industrial design, CAD (SolidWorks, Fusion 360, CATIA), 3D printing, rapid prototype iteration loops, titanium enclosures, mechanical packaging (powertrain, accumulator, motor mounts), ODM sourcing (CES, China), Raspberry Pi, sensors.
Programming	Python, MATLAB / Simulink, Arduino, firmware bring-up (BLE, audio, touch), backend (transcription, retrieval, async LLM).
Operations	Vendor & partner management, user research at scale, GTM (LinkedIn, Reddit, outbound), fundraising (pre-seed), cross-functional leadership.

PUBLICATIONS & PATENT

- Dasadia K., Gandhi M., Banerjee J. *Numerical Validation of Power Take-off Damping in an OWC Chamber and Modifications for Increased Efficiency.* Springer Lecture Notes in Mechanical Engineering, 2022.
- Choukse P., Dasadia K., et al. *Pipe Climbing Robot.* India Design Patent 337173001, 2021.